Duplicate Record System Detection

DATA SCIENCE (24AD3107) Project Review-1

Under the Guidance of
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Introduction

Data is an important resource for organizations, but datasets often contain duplicate or highly similar records. Duplicate records may occur because of repeated data entry, spelling variations, different formatting, missing information, or multiple entries for the same entity.

The Duplicate Record Detection System is a data science-based solution designed to identify duplicate and near-duplicate records automatically. The system analyzes important attributes such as names, email addresses, phone numbers, addresses, and other relevant fields.

The proposed system uses Python, Pandas, and NumPy for data preprocessing and manipulation. It applies normalization, string matching, and similarity-based techniques to compare records and identify duplicates.

The system can be useful for customer databases, student records, healthcare datasets, transaction records, and other real-world datasets where duplicate information can affect data quality and analysis.

Problem Statement

Real-world datasets frequently contain duplicate and near-duplicate records due to repeated data entry, inconsistent formatting, spelling variations, and incomplete information.

These duplicate records can:

- Reduce data quality.
- Cause inaccurate statistical analysis.
- Affect machine learning model performance.
- Increase storage and processing requirements.
- Produce incorrect reports and decision-making results.

Therefore, there is a need for an automated Duplicate Record Detection System that can preprocess datasets and identify exact as well as highly similar records efficiently.

Objectives

- To **preprocess raw datasets** by handling missing values, unnecessary spaces, and inconsistent formats.
- To **normalize data** such as names, email addresses, phone numbers, and addresses.
- To **identify exact duplicate records** within a dataset.
- To **detect near-duplicate records** with minor spelling or formatting variations.
- To apply **string matching and similarity measurement techniques** for comparing records.
- To **automate the duplicate detection process** and reduce manual effort.
- To improve **data accuracy, consistency, and reliability**.
- To generate a **cleaned dataset** by identifying or removing duplicate records.
- To support duplicate detection in **customer, student, healthcare, transaction, and other datasets**.
- To provide reliable and high-quality data for **further analysis and decision-making**.

Literature Survey

	Authors	Title of the Paper	Methodology	Inferences	Limitation / Future Work
1	**Ahmed K. Elmagarmid, Panagiotis G. Ipeirotis, Vassilios S. Verykios (2007)**	**Duplicate Record Detection: A Survey** – *IEEE Transactions on Knowledge and Data Engineering*	Survey of duplicate detection techniques, similarity metrics, approximate matching and record-linkage algorithms.	Provides a comprehensive foundation for identifying duplicate and approximately duplicate records and discusses methods for improving efficiency and scalability.	The paper is primarily a survey; developing efficient techniques for large and complex datasets remains an important research direction.
2	**Mikhail Bilenko,** Raymond J. Mooney (2003)	**Adaptive Duplicate Detection Using Learnable String Similarity Measures** – *ACM SIGKDD*	Uses trainable string-similarity measures and Support Vector Machine-based techniques to learn field-specific similarity.	Demonstrates that learnable similarity measures can improve duplicate-detection accuracy compared with manually selected distance measures.	Training data and domain-specific similarity learning may be required; performance can depend on the quality of training examples.
3	Elizabeth Durham, Yuan Xue, Murat Kantarcioglu, Bradley Malin (2012)	**Quantifying the Correctness, Computational Complexity, and Security of Privacy-Preserving String Comparators for Record Linkage** – *Information Fusion*	Evaluates string comparison methods for privacy-preserving record linkage, considering correctness, computational complexity and security.	Shows the importance of reliable string comparators when matching records while protecting sensitive identifiers.	Privacy-preserving approaches introduce additional computational and security considerations compared with conventional matching.

Problem Gap Identification

The existing duplicate detection approaches have several limitations when applied to real-world datasets.

Identified gaps

1. Exact matching limitations

Exact matching cannot detect records containing minor variations.

2. Inconsistent data formats

Names, phone numbers, addresses, and other fields may have different formats.

3. Spelling variations

Small spelling differences can cause duplicate records to be treated as different records.

4. Unnecessary spaces and formatting

Additional spaces, capitalization, and formatting differences can affect matching.

5. Missing values

Some records may contain incomplete information, making direct comparison difficult.

6. Near-duplicate detection

Identifying records that are similar but not completely identical requires similarity-based techniques.

Proposed solution to the gap

The proposed system combines data preprocessing, normalization, exact matching, string matching, and similarity-based duplicate detection to improve the identification of both exact and near-duplicate records.

Innovation, Creativity, and Novelty

The proposed system focuses on combining multiple data-cleaning and duplicate-detection techniques into a single automated workflow.

Key innovative aspects

- Automated duplicate detection.
- Detection of both **exact and near-duplicate records**.
- Preprocessing before duplicate detection.
- Normalization of inconsistent data.
- Similarity-based comparison of textual attributes.
- Support for multiple attributes such as name, email, phone number, and address.
- Generation of a cleaned dataset for further analysis.
- Can be adapted to different types of real-world datasets.

Example

Instead of considering:

Ravi Kumar

and

RAVI KUMAR

as two different records, preprocessing and normalization can convert them into a consistent format and help identify them as potential duplicates.

Feasibility and Project

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Project Timeline – Duplicate Record Detection System

Phase No.	Phase	Duration (Weeks)	Week 1–2 (1st Month)	Week 3–4 (2nd Month)	Week 5–6 (3rd Month)	Week 7–8 (4th Month)	Week 9–10 (5th Month)	Week 11–12 (6th Month)
1	Requirement Analysis & Planning	2 weeks						
2	Data Collection & Understanding	2 weeks						
3	Data Preprocessing & Cleaning	3 weeks						
4	Data Normalization	2 weeks						
5	Duplicate Detection (Exact & Near)	3 weeks						
6	Model Evaluation & Testing	2 weeks						
7	Final Report & Documentation	2 weeks						
8	Presentation & Review	1 week						

Phase-wise Explanation

1 Requirement Analysis & Planning (2 weeks)

- Understand the problem, define scope, identify datasets and requirements.
- Prepare detailed project plan.

2 Data Collection & Understanding (2 weeks)

- Collect relevant datasets (e.g., customer, student, healthcare, etc.).
- Analyze data structure, attributes and quality issues.

3 Data Preprocessing & Cleaning (3 weeks)

- Handle missing values, remove duplicates, trim spaces, and fix format issues.
- Prepare data for analysis.

4 Data Normalization (2 weeks)

- Standardize names, emails, phone numbers, addresses, etc.
- Convert data to a consistent format.

5 Duplicate Detection (Exact & Near) (3 weeks)

- Apply string matching and similarity measures.
- Identify exact and near-duplicate records.

6 Model Evaluation & Testing (2 weeks)

- Validate results with test data.
- Measure accuracy, precision and recall.

7 Final Report & Documentation (2 weeks)

- Prepare project report, diagrams and results.
- Document code and user guide.

8 Presentation & Review (1 week)

- Create PPT and prepare for review.
- Present results and get feedback.

Challenges

Major challenges expected

1. Handling large datasets efficiently.
2. Managing missing or incomplete values.
3. Handling different data formats.
4. Identifying spelling variations.
5. Selecting suitable similarity thresholds.
6. Detecting near-duplicates accurately.
7. Avoiding false duplicate identification.
8. Maintaining important original information while cleaning data.

Expected Outcome

The proposed system is expected to:

- Identify exact duplicate records.
- Identify highly similar or near-duplicate records.
- Improve dataset consistency.
- Reduce redundant information.
- Generate a cleaned dataset.
- Improve the reliability of data used for further analysis.
- Provide an automated approach to duplicate record detection.

Deploy

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Settings

Upload a CSV

Upload

200MB per file • CSV

Using data/sample_data.csv

Field weights

A design choice, not a universal truth. Fields where two records are unlikely to agree by accident deserve more weight.

Name 0.25

Email 0.25

Phone 0.25

Address 0.15

City 0.10

Weights sum to 1.00. They are renormalized per pair, so this does not have to be 1.00

Duplicate Record Detection System

Finds records that describe the same person even when they are written differently. Pipeline: inspect → normalize → block → compare → score → classify → evaluate.

1. Overview | 2. Preprocessing | 3. Exact duplicates | 4. Blocking | 5. Results | 6. Compare a pair | 7. Evaluation

Records	Columns	Missing values	Exact duplicate rows
445	7	53	13

Why inspect before processing?

- **Data types** - phone and pincode are read as text, not numbers. They are identifiers that happen to be made of digits, and reading them as numbers would drop leading zeros.
- **Missing values** - a field that is often blank makes a weak feature. We need to know that before we trust it.
- **Unique values** - this is what tells us which fields are strong evidence, and it is where the scoring weights come from.

Missing values per column

column	missing	missing_pct
email	14	3.15
phone	14	3.15
pincode	13	2.92
address	12	2.7
record_id	0	0
name	0	0
city	0	0

Unique values per column

uniqueness_ratio = unique values / rows. Close to 1 means the column behaves like an identifier, so a match is strong evidence. Close to 0 means it is categorical, so a match is weak evidence because unrelated records agree by chance.

column	unique_values	uniqueness_ratio	kind	sample
record_id	445	1	identifier-like	R0001, R0002, R0003
name	365	0.82	categorical-like	Madhuri Rathore, Ravi Singh, Shreya Jain
email	379	0.852	categorical-like	mrathore@rediffmail.com, ravi_singh@yahoo.com, sjain@redi
phone	395	0.888	categorical-like	9667600835, 9868011280, 8827109104
address	388	0.872	categorical-like	244 MG Road Howrah, 214 Hill View Road Behala, 56 Main Road
city	28	0.063	categorical-like	Kolkata, Mumbai, Pune
pincode	47	0.106	categorical-like	700029, 400703, 411045

city	0	0		

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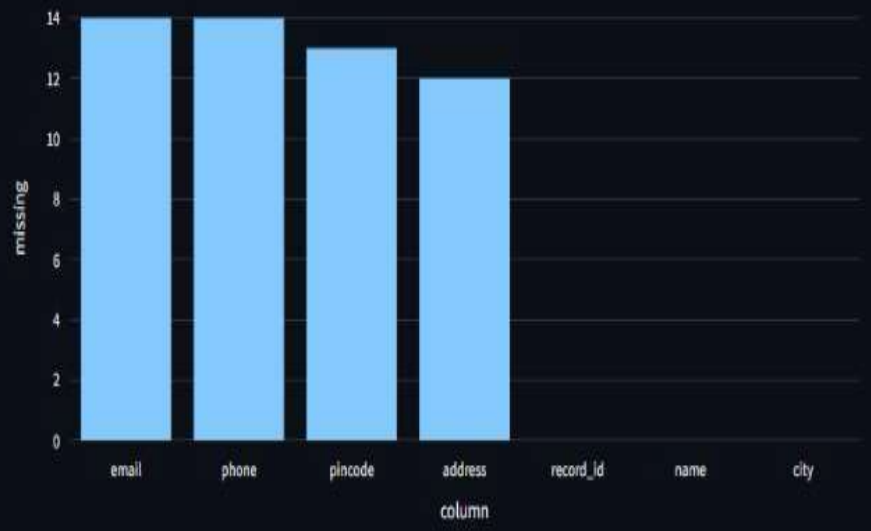
Phone 0.25

Address 0.15

City 0.10

Weights sum to 1.00. They are renormalized per pair, so this does not

Missing values by column



Raw data

record_id	name	email	phone	address	city	pincode
R0001	Madhuri Rathore	mrathore@rediffmail.com	9667600835	244 MG Road Howrah	Kolkata	700029
R0002	Ravi Singh	ravi_singh@yahoo.com	9868011280	214 Hill View Road Behala	Kolkata	700029
R0003	Shreya Jain	sjain@rediffmail.com	8827109104	56 Main Road Andheri	Mumbai	400703
R0004	Yogesh Malhotra	yogeshmalhotra@yahoo.com	8721228160	110 Temple Street Kharadi	Pune	411045
R0005	Dinesh Bhat	dinesh.bhat@outlook.com	9915452267	167 Rose Garden Street Ballygunge	Kolkata	700091
R0006	Kiran Desai	kiran.desai41@outlook.com	6844989266	94 Sector 5 Road Adyar	Chennai	600096
R0007	Kiran S	kiranshetty@yahoo.com		236 ANNA NAGAR BEHALA	CCU	700156

Normalization: removing differences that carry no meaning

Two records about the same person differ in two ways. **Meaningless** differences (capitals, spacing, +91, commas, Rd vs Road) are deleted here. **Meaningful** differences (a genuinely different phone number) are left for the similarity stage to weigh.

Raw columns are never overwritten - each gets a `_norm` twin, so the dashboard can still show what was actually typed.

column	unique_before	unique_after	collapsed	reduction_pct
name	365	329	36	
email	379	372	7	
phone	395	362	33	
address	388	340	48	
city	28	7	21	
pincode	47	47	0	

Distinct values before and after normalization

High confidence

94

Possible

14

Probably unique

999

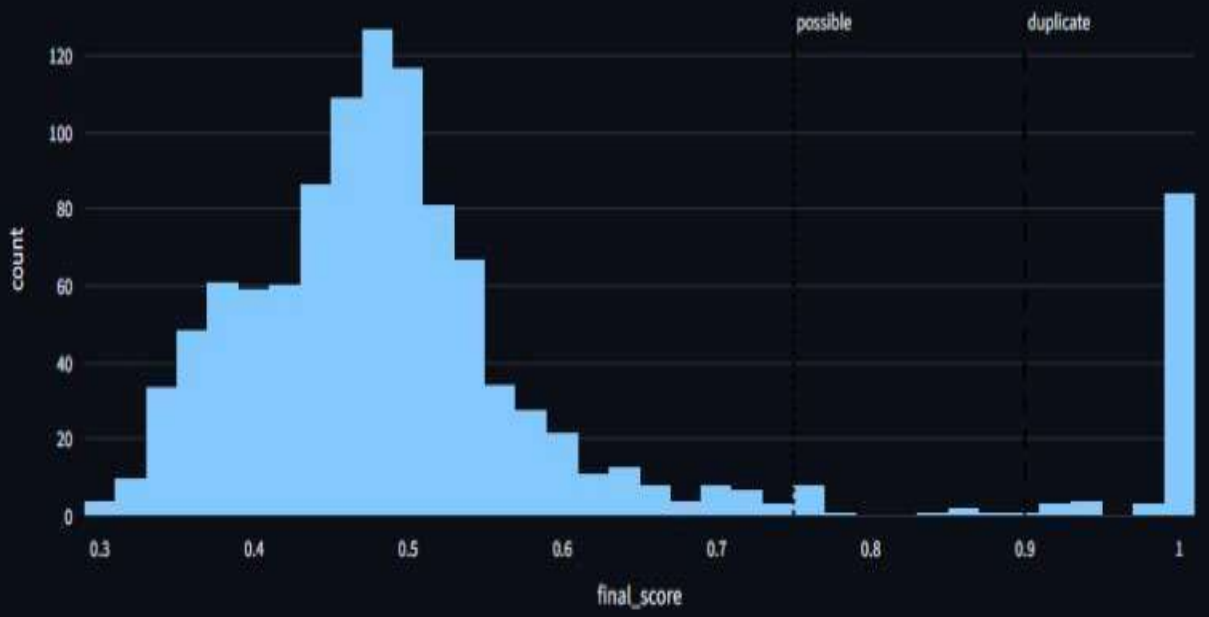
A **similarity score** and a **classification** are different things. The score is evidence - 0.83 means 'fairly similar'. The classification is an action: merge, send for review, or leave alone. Where the cut-offs go depends on what a mistake costs, not on the data.

Here: at or above **0.90** we call it a duplicate; between **0.75** and **0.90** a human should look.

How the candidate pairs were classified



Distribution of match scores



The score histogram is bunched at the two ends: obvious matches near 1.0 and obvious non-matches low down. The thin middle is where every mistake lives, and it is what the review band exists for.

Conclusion

The **Duplicate Record Detection System** aims to address the problem of duplicate and inconsistent records in real-world datasets. The proposed approach combines data preprocessing, normalization, exact matching, string matching, and similarity-based techniques.

By automatically identifying duplicate and near-duplicate records, the system can improve **data accuracy, consistency, and reliability**. The cleaned dataset can subsequently be used for more reliable analysis and machine learning applications.

Future Work

The system can be further enhanced by:
Supporting larger datasets.
Improving similarity detection techniques.
Using machine learning models for duplicate classification.
Automatically selecting similarity thresholds.
Supporting additional data types and attributes.
Developing a user-friendly web interface.
Providing duplicate-detection reports and visualizations.
Integrating the system with databases and data-management platforms.

References

1. Python Documentation – Python Programming Language.
2. Pandas Documentation – Data Analysis and Data Manipulation.
3. NumPy Documentation – Numerical Computing with Python.
4. Documentation and research resources related to data preprocessing.
5. Research literature related to record linkage, entity resolution, and duplicate detection.

Publications Details

Title: *Duplicate Record Detection Using Attribute Normalisation, Blocking and Similarity Fusion*
Authors: J. Kumudavalli, R. Varshitha, P. Abhinaya
Affiliation: Department of Computer Science and Engineering, KL University, KLH Bowrampet Campus, Hyderabad, India
Research Area: Data Science, Data Cleaning & Entity Resolution
Keywords: Duplicate Record Detection, Entity Resolution, Data Cleaning, Blocking, String Similarity, Data Quality
Publication Format: IEEE-style research paper draft